

YUDHA STYAWAN

Lecturer, Geophysicist & Computational Seismology Enthusiast

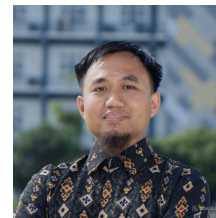
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🎓 Google Scholar: [s.itera.id/yudhascholar](https://scholar.google.com/citations?user=s.itera.id/yudhascholar)

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🐙 GitHub: github.com/yudhastyawan



📁 PROFESSIONAL EXPERIENCE

Secretary at Earthquake and Tsunami Disaster Mitigation Center

2025 - Present

Institut Teknologi Sumatera, Indonesia

Assisting in the management and coordination of center activities, including research programs, public outreach, and administrative duties related to disaster mitigation.

Lecturer - Geophysical Engineering

2022 - Present

Institut Teknologi Sumatera, Indonesia

Teaching undergraduate (B.Sc.) courses in geophysics. Mentoring students in practical applications and research projects. Teaching focus areas: seismology, geodynamics, seismic data analysis, engineering seismology, and computational modeling.

Laboratory Staff - Geophysical Engineering

2018 - 2021

Institut Teknologi Sumatera, Indonesia

Managing laboratory equipment, assisting students during practical sessions, and maintaining geophysical instruments for field measurements.

🎓 EDUCATION

Master of Science (M.Sc.) in Geophysics

2021

National Central University, Taiwan

Thesis: Characteristics of Seismic Attenuation in Sumatra Subduction Zone, Indonesia

Advisors: Asst. Prof. Chun-Hsiang Kuo, Prof. Bor-Shouh Huang

Bachelor of Engineering (S.T.) in Geophysical Engineering

2018

Institut Teknologi Sumatera, Indonesia

Thesis: Lindu Software: Aplikasi Pengolahan Data Seismologi Berbasis Python untuk Tomografi Waktu Tempuh [Lindu Software: Python-Based Seismological Data Processing Application for Travel Time Tomography]

Advisors: Dr. Tedi Yudistira, S.Si., M.Si., Ruhul Firdaus, S.T., M.T.

💰 GRANTS & RESEARCH FUNDING

ITERA Expertise-Based Research Grant

2025

Institut Teknologi Sumatera (ITERA)

Application of Brownian Passage Time Method as Recurrence Interval Calculation on Fault Earthquakes in the Western Part of Sunda-Java Strait to Support the Update of Seismic Hazard Assessment Model in Lampung Region

ITERA Assignment Research Grant

2025

Institut Teknologi Sumatera (ITERA)

Policy Brief Response of Lampung Province City and Regency Towards Megathrust

ITERA Scientific Group Strengthening Research Grant

2025

Institut Teknologi Sumatera (ITERA)

Azimuth Variation Analysis on Single Station HVSR Measurement in Umbul Niti Geothermal Manifestation, Jatimulyo Village, South Lampung Regency

ITERA Community Service Funding

2025

Institut Teknologi Sumatera (ITERA)

Early Preparedness: Forming a Tsunami Responsive Generation in Coastal Schools

ITERA Beginner Lecturer Research Grant

2024

Institut Teknologi Sumatera (ITERA)

Updating Seismic Activity Modeling and Vs30 on Ground Motion Prediction Equations for Long-term Seismic Hazard Assessment in Sumatra, Indonesia: A Probabilistic Approach

RECENT PUBLICATIONS

- Alif, S. M., Anggara, O., Perdana, R. S., Wulandari, R., & **Styawan, Y.** (2026). Earthquake potential and seismic hazard in Southern Sumatra, Indonesia: Insights from different GNSS velocity sets. *Journal of Asian Earth Sciences*, 311, 107217. <https://doi.org/10.1016/j.jseaes.2026.107217>
- Antosia, R. M., Endryanti, W. M., Farduwin, A., Rizki, R., & **Styawan, Y.** (2026). Soil characterization for seismic vulnerability and preliminary subsidence assessment based on microtremor analysis in Sukarame District, Bandar Lampung. *Journal of Degraded and Mining Lands Management*, 13(3), 10439-10450. <https://doi.org/10.15243/jdmlm.2026.133.10439>
- Junian, W. E., **Styawan, Y.**, Prasetyo, N., & Paembonan, A. Y. (2026). Applying the one-to-one-based optimizer (OOBO) algorithm for one-dimensional inversion modeling of magnetotellurics data. *Pure and Applied Geophysics*, 183(6), 2873-2890. <https://doi.org/10.1007/s00024-026-03998-x>
- Wulandari, R., Suhendi, C., & **Styawan, Y.** (2026). Receiver-fault variability and depth-dependent Coulomb stress changes in the 2019 Mw 7.2 Halmahera earthquake sequence. *Jurnal Gecelebes*, 145-159. <https://doi.org/10.70561/gecelebes.v10i2.47046>
- Wulandari, R., **Styawan, Y.**, & Chan, C.-H. (2026). Enhancing seismic hazard preparation in Lampung, Sumatra: Improved magnitude conversion, seismicity smoothing, and area source modeling. *Indonesian Journal on Geoscience*, 13(2), 221-241. <https://doi.org/10.17014/ijog.13.2.221-241>
- **Styawan, Y.** (2025). Optimizing seismic b-values in the Java region through Voronoi-based OK1993 modelling. *JGE (Jurnal Geofisika Eksplorasi)*, 11(2), 109-121. <https://doi.org/10.23960/jge.v11i2.489>
- **Styawan, Y.** (2025). QuakeSee: Aplikasi cross-platform Python berbasis web untuk otomasi dan aksesibilitas dalam pengunduhan data gempa terbuka [QuakeSee: A cross-platform web-based Python application for automation and accessibility in downloading open earthquake data]. *GeoScienceEd Journal*, 6(3), 1292-1301. <https://doi.org/10.29303/goescienceed.v6i3.968>
- Farduwin, A., Nugraha, P. N., **Styawan, Y.**, Lestari, E. Y. P., & Tr, D. P. J. (2025). Site effects identification using HVSR method in Cisarua hot spring area, Natar, South Lampung. *JGE (Jurnal Geofisika Eksplorasi)*, 11(2), 151-162. <https://doi.org/10.23960/jge.v11i2.494>
- Hamidah, I. F., Farduwin, A., **Styawan, Y.**, Nurfitriani, I., Prasetyo, N., Junian, W. E., & Wulandari, R. (2025). Analisis ancaman gempa Lombok menggunakan metode spasial temporal a-value dan b-value periode 1964-2022 [Lombok earthquake hazard analysis using spatio-temporal a-value and b-value method for the period of 1964-2022]. *Wahana Fisika*, 10(1), 12-26. <https://doi.org/10.17509/wafi.v10i1.76470>

</> OPEN SOURCE SOFTWARE

- **SeisBox** (<https://github.com/yudhastyawan/SeisBox>): A Rust-based native desktop app for downloading open-source catalog and seismogram data, visualizing waveforms, picking seismic wave phases, computing Coulomb stress changes, analyzing HVSr, and performing 1D Vs inversions.
- **QuakeSee** (<https://github.com/yudhastyawan/quakesee>): A Python GUI program for downloading and interactively visualizing open-source seismic data to aid undergraduate seismology practicums.
- **SeisWave** (<https://github.com/yudhastyawan/seiswave>): A robust Python and Fortran framework for surface wave modeling and Full f-c Spectrum Inversion, utilizing dispersion image code from Computer Programs in Seismology.
- **Lindu Software** (<https://github.com/Computation-Geophysics-TG-Itera/lindu-software>): An open-source Python GUI tool for earthquake location, HypoDD relocation, and traveltimes tomography, though its development is currently paused.

TECHNICAL SKILLS

Programming & Dev	Python, Rust, Julia, Fortran, C++ PyQt, Git, GMT, Bash / CLI, Linux, LaTeX
Geophysical Tools	Seismometers (Primary); Gravity, Magnetic, and Electrical instruments
Professional Interests	Computational Seismology, Engineering Seismology, Geophysics Software Development